Reading a Behavioral Code Across Mice, Days, and Model Classes: Longitudinal Leave-One-Animal-Out Decoding of Mesoscale Widefield Calcium Imaging

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Abstract

Decoding behavior from cortex-wide widefield calcium imaging is now routine, but almost always within a single animal. Whether a cortex-wide behavioral decoder transfers to an unseen individual—and, if so, whether that transfer is stable over days and robust to model capacity—has not been characterized. We study this on the open BraiDyn-BC cohort (25 mice, a 15-day cued lever-pull protocol; dorsal-cortex $\Delta F/F$ parcellated into 44 Allen-atlas regions), decoding four discrete task events (lever-pull, lick, reward, tone) with a strict leave-one-mouse-out (LOMO) protocol in the shared parcel space. A simple, interpretable linear decoder trained on $N-1$ mice reads every held-out mouse above chance for every event (LOMO AUC 0.80–0.93, permutation $p = 0.007$), landing within 0.03–0.04 AUC of the within-animal ceiling—a small residual cross-subject gap. Our contributions are what this gap does, not merely that it is small. (i) Longitudinal stability: exploiting the 15-day protocol, a decoder trained on other mice's first-week sessions reads a held-out mouse's last-week sessions with no loss on any of the four events, and within-mouse accuracy is flat in the train–test day-gap (≤ 0.038 AUC over two weeks)—to our knowledge the first longitudinal characterization of a cross-animal cortical decoder. Within a single animal the same design does resolve a small drift on the strongest event (≈ 0.02 AUC) that the cross-animal readout does not inherit. (ii) Capacity-invariance: nonlinear temporal models (a 1-D CNN and a GRU over the evoked window) raise cross-animal AUC by +0.02 to +0.12— markedly for the two self-initiated movements, only slightly for the two cue-locked events—while the cross-subject gap stays within a narrow 0.004–0.047 band and is smaller than the linear model's in seven of eight comparisons, so the shared structure includes the response dynamics, not only the linear evoked mean. (iii) Replication and interpretability: the effect reproduces on an independent widefield cohort (different lab, atlas, and task), and decoder weights localize to a shared midline retrosplenial/visual-association backbone with target-specific somatosensory and motor modulation. We position these results relative to recently published atlas-aligned foundation-model work, which they complement with an interpretable, near-ceiling, longitudinally-controlled baseline. Code and a streamed-feature pipeline (no raw-video download) are released.

1 Introduction

Mesoscale (widefield) calcium imaging records activity across the dorsal cortical surface during behavior, and cortex-wide dynamics are strongly modulated by movement and task events (Musall et al., 2019; MacDowell & Buschman, 2020). Decoding behavior from this signal is now standard practice, but the overwhelming majority of studies train and evaluate within a single animal: a model is fit on some trials of one mouse and tested on other trials of the same mouse. This answers "can we read this brain?" It does not answer whether the cortex-wide behavioral code is the same across individuals—a question that determines whether a shared, atlas-referenced representation exists or whether each brain is idiosyncratic—nor whether any such shared readout is stable over days or robust to the decoder's capacity.

Two facts make the cross-animal question non-trivial. First, vasculature, hemodynamic coupling, and atlas registration differ between mice, and these nuisances could dominate the parcellated signal. Second, even if a shared code exists at one moment, cortical representations are known to drift over days at the single-neuron level (Rule et al., 2020), so a decoder that transfers across mice on day 1 need not transfer on day 15. The recently released BraiDyn-BC resource (Kondo et al., 2025) is unusually well suited to settle both: it provides cortex-wide $\Delta F/F$ parcellated into a shared 44-region Allen atlas from 25 mice, each recorded across a ∼15-session, multi-week operant protocol. Its data descriptor performs no decoding of any kind, leaving these questions open on the cohort.

We give a focused, interpretable answer. Working entirely in the shared 44-parcel space—so that "parcel j" means the same anatomical region in every mouse and no per-subject alignment is needed—we perform strict leave-one-mouse-out (LOMO) decoding of four discrete task events, and then ask what happens to cross-animal readout across time and across model capacity.

Contributions.

- An interpretable near-ceiling cross-subject baseline. A 44-dimensional linear LOMO decoder reads every held-out mouse above chance for all four events (LOMO AUC 0.80–0.93, $p = 0.007$), within 0.03–0.04 AUC of the within-animal ceiling. This is a small, honestly reported residual gap—not a claim of zero gap.
- Longitudinal stability (primary result). Using the 15-day protocol, we show the cross-animal decoder does not degrade over time on any of the four events: trained on other mice's early sessions (days 1–5), it reads a held-out mouse's late sessions (days 11–15) with no loss, and within-mouse accuracy is essentially flat in the train–test day-gap. We are not aware of a prior longitudinal characterization of a cross-animal cortical decoder.
- Capacity-invariance. Nonlinear temporal decoders (1-D CNN, GRU) improve cross-animal AUC without systematically widening the cross-subject gap—it is in fact smaller than the linear model's in seven of eight comparisons—indicating the shared component of the code encompasses the temporal response and not merely its collapsed mean. The size of the gain dissociates by event type: large for self-initiated movements, small for cue-locked events.
- Replication + interpretability. The pattern reproduces on an independent widefield cohort, and the decoders localize to interpretable, partly target-specific cortex.

We are explicit (Section 2) that cross-animal conservation of cortex-wide dynamics is already established and that a recently published foundation model performs atlas-aligned cross-subject decoding on this dataset; our aim is the complementary, interpretable, and longitudinally-controlled characterization those works do not provide.

2 Related work

Cross-subject widefield modeling. Most directly related, WiCAT (Hosseini et al., 2026) learns atlas-aligned, subject-invariant representations of widefield imaging and performs zero-shot cross-subject behavior decoding on the same Kondo/BraiDyn-BC cohort (25 mice, 352 sessions), alongside a second (Musall) cohort, using self-supervised pretraining over spatiotemporal tokens registered to the Allen CCF without session-specific components. It establishes that cross-subject widefield decoding is achievable, and we accordingly claim no priority on that concept. Our contribution is complementary along three concrete axes. (a) Representation. We decode in the interpretable 44-parcel atlas space with a simple linear/CNN/GRU LOMO protocol rather than a learned-token foundation model, so the fitted weights are directly readable as cortical anatomy (Section 4.4). (b) Target and metric. WiCAT decodes continuous behavior on held-out subjects (reported zero-shot

Kondo $R^2 = 0.18$ under linear probing, 0.29 with a Kalman decoder), whereas we decode discrete task events and report AUC. These quantities measure different prediction problems under different metrics and are not directly comparable; we make no claim of outperforming WiCAT. Read together they are nonetheless suggestive—near-ceiling cross-animal AUC on discrete events alongside modest zero-shot R^2 on continuous trajectories hints that the individual-invariant component of the code carries event structure more readily than moment-to-moment behavioral detail—though because architecture, supervision, and metric all differ simultaneously, we offer this as a conjecture for future controlled work rather than a result. (c) Uncharacterized axes. Neither longitudinal stability nor capacity-invariance of cross-subject transfer is examined there; those are our contributions (i) and (ii).

Conservation of cortex-wide dynamics. That cortex-wide activity is shared across individuals is established. MacDowell & Buschman (2020); MacDowell et al. (2024) show a small set of cortex-wide spatiotemporal "motifs" explains $> 75\%$ of variance and generalizes across mice, with animals differing mainly in how often shared motifs are expressed; Musall et al. (2019) show cortex-wide single-trial activity is dominated by movement. These are unsupervised decompositions and within-animal encoding analyses; they motivate our premise but do not decode behavior across animals. Our conservation-adjacent result (the small LOMO-vs-within gap) is thus a confirmation of this consensus, not a discovery—which is why we foreground stability and capacity instead.

Cross-subject decoding in other modalities. Training a decoder on one animal and testing on another is established in electrophysiology via manifold alignment in rat sensorimotor cortex (Melbaum et al., 2022) and via contrastive embeddings that expose a "universal" hippocampal code (Schneider et al., 2023). These use different modalities/species and typically alignment-based matching rather than an alignment-free, shared-atlas LOMO decoder, and none address multi-day stability of the transfer.

Deep decoders on widefield imaging. End-to-end CNN–RNN models decode behavioral state from cortex-wide calcium imaging and outperform linear baselines (Ajioka et al., 2024), but for within-animal splits, pixel-level input, and locomotion rather than discrete events. Thus that nonlinear beats linear is known within animals; whether capacity changes cross-subject transfer—our question (ii)—is open.

3 Data and methods

Dataset. BraiDyn-BC (Kondo et al., 2025) (DANDI:001425, CC-BY) provides, per mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ parcellated into 44 Allen Common Coordinate Framework regions (22/hemisphere) at $\sim$30 Hz, with time-aligned behavioral channels (lever state, lick, reward, tone). Twenty-five mice each performed $\sim$15 daily operant sessions spanning $\sim$3 weeks. Raw movies are large ($\sim$5 GB/file) but the parcellated traces are tiny, so we stream only the 44-region $\Delta F/F$ and behavioral channels from cloud storage, never downloading raw imaging.

Events and features. For each target we detect event onsets as rising edges (1 s refractory). On the operant task sessions all four channels are well populated (reward fires on each rewarded trial, tone on each cue), so all four are decoded. For each onset we form a 44-dimensional evoked feature: mean post-onset (0–1 s) parcellated $\Delta F/F$ minus a pre-onset (-0.5–0 s) baseline. Negative examples are matched-count windows drawn > 2 s from any event. The cross-animal analysis pools three task sessions per mouse spread across the protocol.

Decoding and evaluation. The core decoder is a standardized ℓ_2-regularized logistic regression. The evaluation axis is leave-one-mouse-out (LOMO): train on all events of $N-1$ mice, test on the held-out mouse; the 44 atlas parcels are shared across mice, so the split needs no per-subject alignment. We report mean held-out-mouse AUC, a cluster (over-mice) bootstrap 95% CI, and a permutation null in which event/baseline labels are shuffled within each mouse before re-running the full LOMO pipeline (150 permutations). As a positive control

Table 1: Cross-mouse (leave-one-mouse-out) decoding of four task events. Every held-out mouse is above chance for every target; the cross-animal AUC sits 0.03–0.04 below the within-mouse ceiling.

Target	Mice	Within-mouse AUC	LOMO AUC (95% CI)	p
Lever-pull	25	0.842	0.804 $[0.780, 0.829]$	0.007
Lick	25	0.885	0.850 $[0.822, 0.874]$	0.007
Reward	25	0.953	0.926 $[0.911, 0.938]$	0.007
Tone	25	0.864	0.828 $[0.797, 0.859]$	0.007

we run within-mouse 5-fold cross-validation. Per-parcel importance is the magnitude of the standardized logistic weights.

Longitudinal design. To probe temporal stability we pool each mouse's daily task sessions into an early block (days 1–5) and a late block (days 11–15) and cross the within/across-animal axis with the same/across-block axis, giving four regimes: within-mouse same-block (5-fold CV on early; WS), within-mouse across-block (train early → test late, same mouse; WX), LOMO same-block (train other mice's early → test held-out early; LS), and LOMO across-block (train other mice's early → test held-out mouse's late; LX). The across-block change is tested by a sign-flip permutation test on the per-mouse paired differences. A within-mouse train-day/test-day sweep over all ordered session pairs yields the accuracy-vs-day-gap curve and its slope.

Nonlinear temporal decoders. To test whether transfer depends on model capacity we retain the full spatiotemporal window per event (-0.5 to $+1.0$ s, 45×44) and compare the linear evoked-vector model against a 1-D CNN over time (two Conv1d–BN–ReLU blocks, global average pool, dropout, linear head) and a single-layer GRU over the frame sequence, trained with Adam (BCE loss, weight decay, per-channel train-set standardization) under the same within-mouse and LOMO splits.

4 Results

4.1 An interpretable cross-subject baseline: a small residual gap

All four task events decode from the cortex-wide response of a never-seen mouse well above chance (Table 1, Fig. 1): LOMO AUC ranges 0.804 (lever-pull) to 0.926 (reward), all at permutation $p = 0.007$ (the resolution floor of 150 permutations), with bootstrap lower bounds ≥ 0.78. Every one of the 25 held-out mice exceeds chance for every target, with no per-mouse AUC below 0.65 (minima 0.653 lever-pull, 0.661 lick, 0.811 reward, 0.697 tone; Fig. 4, bottom)—these are point estimates, as we do not run a per-mouse significance test. Cross-animal decoding lands within 0.03–0.04 AUC of the within-animal ceiling for all four events (e.g. lever-pull 0.804 vs. 0.842; reward 0.926 vs. 0.953). Because the pooled cross-animal decoder is estimated from $\sim$24$\times$ more data than any single-mouse model, and the within-mouse controls here are themselves well powered (three sessions/mouse), this small gap does not reflect a data imbalance: the behavioral code is largely shared across individuals in the atlas space. We treat this as an interpretable baseline, not the headline; the substantive results are how the gap behaves over time and capacity.

4.2 Longitudinal stability of the cross-animal decoder

Does the shared readout survive the passage of days? Using the four-regime design (Table 2, Fig. 2), run on all four targets, the cross-animal answer is yes; the within-animal picture is more nuanced, and we report both.

Within-animal. Training on days 1–5 and testing on days 11–15 of the same mouse leaves lever-pull marginally better (WX 0.838 vs. WS 0.826) but costs a little on the other three (lick 0.865 vs. 0.883; tone 0.847 vs. 0.860; reward 0.939 vs. 0.961). For lick and tone the bootstrap CIs overlap, but for reward they do not (0.939 $[0.925, 0.952]$ vs. 0.961 $[0.955, 0.967]$):

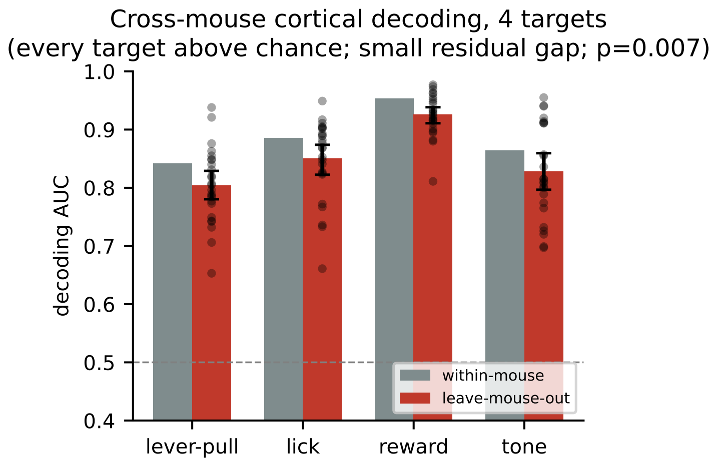

Figure 1: Within-mouse vs. leave-one-mouse-out AUC for all four targets (bootstrap CI; dots = held-out mice). Cross-animal decoding approaches the within-animal ceiling—a small residual gap.

on the cohort's strongest target there is a small but resolvable within-animal drift of ≈ 0.02 AUC over two weeks. We therefore do not claim that within-animal drift is absent—only that it is small, and that it does not propagate to the cross-animal readout.

Cross-animal. Trained only on other mice's first-week sessions and tested on a held-out mouse's last-week sessions (LX), the decoder equals or exceeds its same-block control (LS) for every target: lever-pull 0.818 vs. 0.792 (paired $\Delta = +0.026$, sign-flip $p = 0.009$, $n = 25$), lick 0.842 vs. 0.832 (+0.010, $p = 0.44$), reward 0.929 vs. 0.919 (+0.010, $p = 0.29$), and tone 0.823 vs. 0.813 (+0.010, $p = 0.55$). Only lever-pull is individually significant, so what we claim is the conjunction—all four differences are positive and none is negative—not an improvement. The mildly positive direction is consistent with late-week behavior being more practiced and thus slightly cleaner to decode.

Where the two axes meet is informative: the cross-subject gap narrows on late data exactly where within-animal drift is resolvable. For reward the within-minus-LOMO gap is 0.042 on early sessions (0.961 vs. 0.919) but 0.010 on late ones (0.939 vs. 0.929)—a mouse's own early-week decoder transfers to its own late-week data worse than a decoder pooled over the other 24 mice does. This is consistent with the pooled model averaging out individual drift, but we have not tested that mechanism and report the observation only.

Finally, a within-mouse train-day/test-day sweep ($>$ 2300 session pairs per target) shows accuracy is flat in the day-gap: slopes are -0.0004 (lever-pull), -0.0022 (lick), -0.0025 (reward) and -0.0027 (tone) AUC/day—at most 0.038 AUC across the entire two-week span (Fig. 2, right).

4.3 Capacity-invariance: the gap does not grow with model capacity

The linear baseline collapses each event to one evoked-difference vector, discarding temporal dynamics. A higher-capacity model could exploit those dynamics—but it could also overfit mouse-specific idiosyncrasies and open a cross-animal gap. We test both by comparing the linear model to a 1-D CNN and a GRU over the full window on identical events and splits, for all four targets (Table 3, Fig. 3).

Temporal models raise absolute accuracy, but by an amount that depends sharply on which event is decoded. The gain in cross-animal AUC over the linear model is large for the two movement events—lever-pull $0.804 \to 0.894/0.910$ (CNN/GRU) and lick $0.850 \to 0.966/0.967$, i.e. +0.09 to +0.12—but modest for the two cue-locked events: reward $0.925 \to 0.944/0.958$ (+0.019/+0.033) and tone $0.828 \to 0.864/0.881$ (+0.036/+0.053). Across all eight the range

Table 2: Longitudinal design: within/across-animal × same/across-block AUC, all four targets, $n = 25$ mice. Training on other mice's days 1–5 and testing a held-out mouse's days 11–15 (LX) costs nothing relative to the same-block control (LS) for any target. Within a mouse (WX vs. WS) three of four targets dip slightly, and for reward the dip is resolvable. Δ is the paired LX−LS difference (sign-flip permutation).

	within-mouse		leave-one-mouse-out			
Target	same (WS)	early→late (WX)	same (LS)	early→late (LX)	Δ	p
Lever-pull	0.826	0.838	0.792	0.818	+0.026	0.009
Lick	0.883	0.865	0.832	0.842	+0.010	0.44
Reward	0.961	0.939	0.919	0.929	+0.010	0.29
Tone	0.860	0.847	0.813	0.823	+0.010	0.55

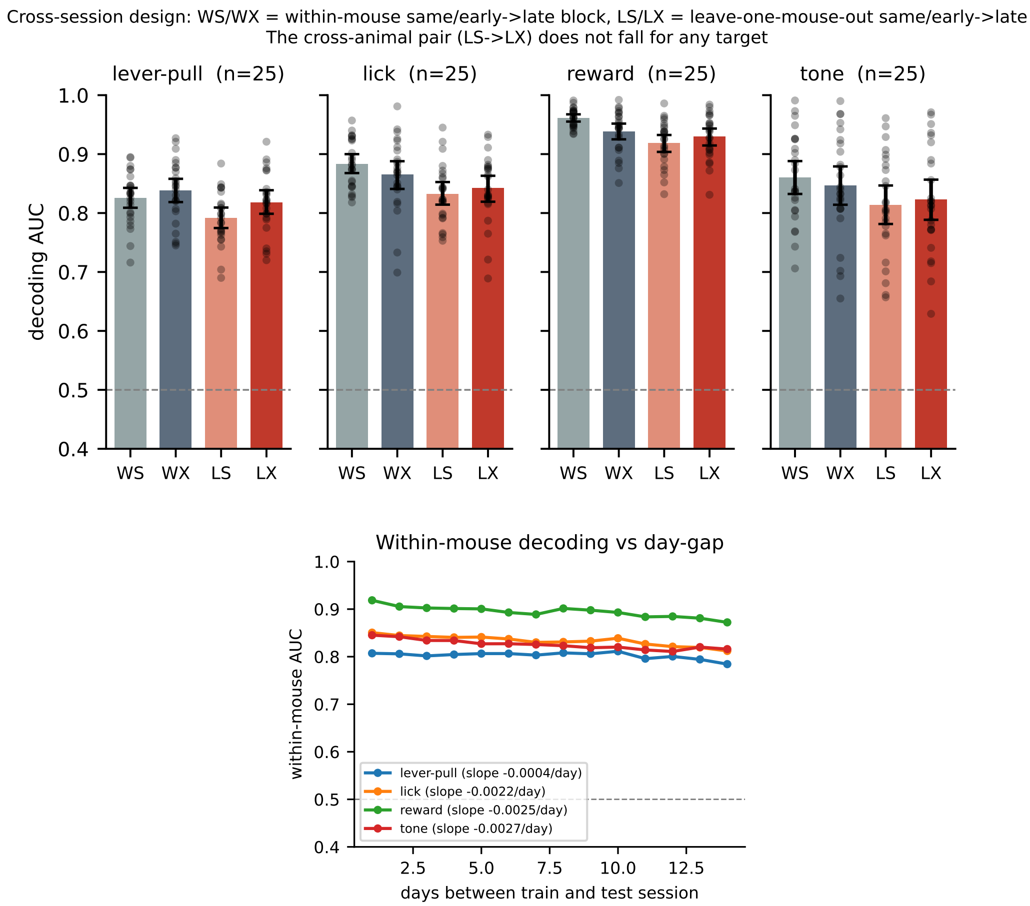

Figure 2: Top: within/across-animal × same/across-block decoding for all four targets (bootstrap CI; dots = mice). The cross-animal pair (LOMO same-block vs. LOMO cross-block) does not fall for any target; the within-mouse pair dips for reward. Bottom: within-mouse accuracy vs. train–test day-gap is flat for every target (slopes −0.0004 to −0.0027 AUC/day).

is +0.019 to +0.117. The dissociation is interpretable—reward and tone are triggered by discrete external cues and evoke stereotyped responses that a collapsed evoked mean already captures well, whereas self-initiated movements carry richer temporal structure that the collapsed vector discards—but we did not predict it in advance and report it as an observation rather than a tested claim.

Table 3: Linear vs. nonlinear temporal decoders, all four targets, $n = 25$ mice. Temporal models are more accurate, yet the cross-subject gap (LOMO − within) stays within 0.004–0.047 and does not systematically exceed the linear model's: it is smaller in seven of the eight nonlinear-vs-linear comparisons, larger only for the lever-pull CNN (by 0.010). Within-mouse and LOMO values are separate runs from the linear entries of Table 1, so they can differ in the third decimal.

Target	Model	Within-mouse	LOMO	Gap
Lever-pull	Linear (evoked)	0.841	0.804	−0.037
	1-D CNN (temporal)	0.942	0.894	−0.047
	GRU (temporal)	0.934	0.910	−0.024
Lick	Linear (evoked)	0.885	0.850	−0.036
	1-D CNN (temporal)	0.978	0.966	−0.012
	GRU (temporal)	0.974	0.967	−0.008
Reward	Linear (evoked)	0.951	0.925	−0.026
	1-D CNN (temporal)	0.962	0.944	−0.018
	GRU (temporal)	0.962	0.958	−0.004
Tone	Linear (evoked)	0.864	0.828	−0.036
	1-D CNN (temporal)	0.886	0.864	−0.022
	GRU (temporal)	0.896	0.881	−0.015

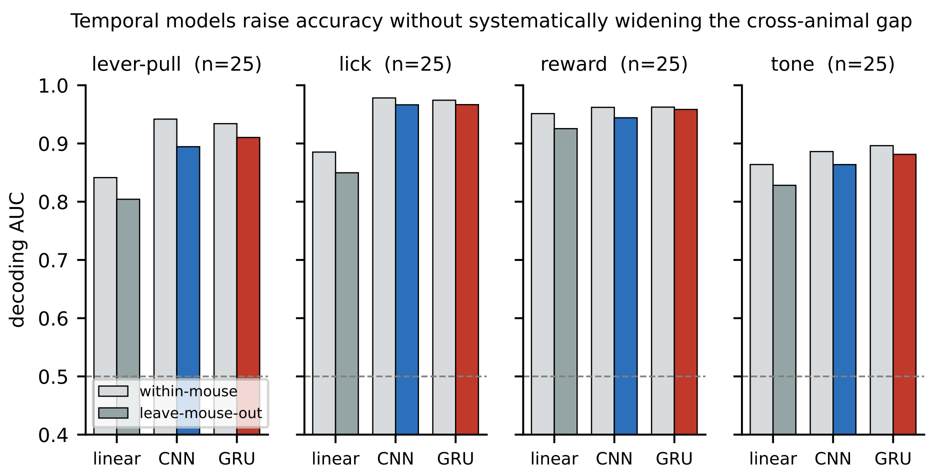

Figure 3: Within-mouse vs. LOMO AUC for the linear and two nonlinear temporal decoders, all four targets. Capacity raises accuracy—markedly for the movement events, only slightly for the cue-locked ones—without opening a cross-animal gap.

Crucially, the cross-subject gap does not grow systematically with capacity. Across all twelve model×target combinations it stays within a narrow 0.004–0.047 AUC band, and in seven of the eight nonlinear-vs-linear comparisons it is smaller than the linear model's—most clearly reward GRU (−0.004 vs. linear −0.026) and lick GRU (−0.008 vs. −0.036). The single exception is the lever-pull CNN (−0.047 vs. linear −0.037); we report this 0.010 excess for completeness but do not interpret it, as we run no test comparing one model's gap against another's. The shared component of the code therefore includes the temporal response and not merely its collapsed mean, and capacity buys accuracy without a systematic cost in transfer.

4.4 Cross-dataset replication and cortical localization

Replication. To test dataset-specificity we repeat the analysis on an independent widefield cohort from a different laboratory (Cardin–Higley; Benisty et al., 2024; 6 mice, 23 CCFv3 parcels, spontaneous behavior, different targets, 6005 windows). Movement (locomotion) reaches LOMO AUC 0.731 (within-mouse 0.744) and arousal (pupil) 0.627 (within 0.645): the same near-ceiling cross-animal readout (gaps 0.014 and 0.018) appears across a different lab, atlas, task, and behavioral variable, indicating a general property of cortex rather than a cohort artifact. The per-animal spread is wider here than on BraiDyn-BC: movement ranges 0.570–0.905 and arousal 0.512–0.726 over the six held-out animals, so the weakest held-out animal sits at chance for arousal. We therefore offer this cohort as evidence that the small cross-animal gap generalizes, not as an equally strong per-animal readout.

Localization. Decoder weights (Fig. 4) are spatially structured. Ranking parcels by standardized-weight magnitude, exactly three of the 44 appear in the top eight of all four targets: retrosplenial agranular cortex (RSPagl, left) and anteromedial visual-association cortex (VISam, bilaterally)—a compact midline retrosplenial/visual-association backbone, as expected when a movement/arousal-related component pervades widefield activity (Musall et al., 2019). Target identity modulates the remainder: the tone (auditory-cue) decoder uniquely recruits nose and upper-limb somatosensory cortex (SSp_n, SSp_ul) together with motor cortex (MOp); the lick decoder recruits lower-limb somatosensory cortex bilaterally (SSp_ll); and the reward decoder recruits neither, its top parcels being the shared backbone plus posterior parietal (VISa), dorsal retrosplenial (RSPd), and barrel cortex (SSp_bfd). The codes thus carry target-specific structure over a shared backbone rather than being pure generic arousal—though we note the somatosensory subfields that separate the targets are not the ones a naive somatotopy of each action would predict, so we report the dissociation without interpreting its anatomy. Parcel names are read from the dataset's own ROI table and released as parcel_names.json.

5 Discussion

On BraiDyn-BC, cortex-wide representations of all four task events generalize to unseen individuals within 0.03–0.04 AUC of within-animal decoding, and—our main contribution—that small gap is stable across a 15-day protocol for every event and does not widen with model capacity for any of them. The one place time is resolvable is within an animal (reward, ≈ 0.02 AUC), and notably the cross-animal readout does not inherit it. The shared component of the code is therefore invariant to individual identity, to session, and to whether the decoder is a 44-parameter linear map or a temporal network; and it includes the response dynamics, not only the collapsed evoked mean. That cortex-wide dynamics are conserved across individuals is not itself new (MacDowell & Buschman, 2020; MacDowell et al., 2024; Musall et al., 2019), and a recently published foundation model performs atlas-aligned cross-subject decoding on this cohort (Hosseini et al., 2026); our contribution is the interpretable, near-ceiling, and longitudinally- and capacity-controlled characterization those lines do not provide.

Limitations. We decode the four thresholdable task-event channels; continuous or graded behavioral variables are a natural next target, and are precisely where the cross-subject problem appears hardest (Hosseini et al., 2026). Our conclusions rest on LOMO-vs-within and across-vs-same-block comparisons rather than absolute AUC, and they hold for all four events and for both the linear and the nonlinear decoders. The movement/cue dissociation in how much capacity buys (Section 4.3) is an unplanned observation on four events, not a tested hypothesis. The residual cross-animal gap is small but non-zero: we characterize rather than eliminate it. The longitudinal-novelty framing is, to the best of our literature search, unaddressed by prior work, but necessarily rests on absence of evidence. Finally, the drift literature we invoke as motivation (Rule et al., 2020) documents reorganization at the single-neuron level, whereas our signal is a mesoscale average over large populations within each parcel; such averaging may attenuate single-cell drift, so our stability result should be read as establishing that the parcel-level code a cross-animal decoder actually consumes is stable—not as evidence against single-neuron drift, which we cannot observe here.

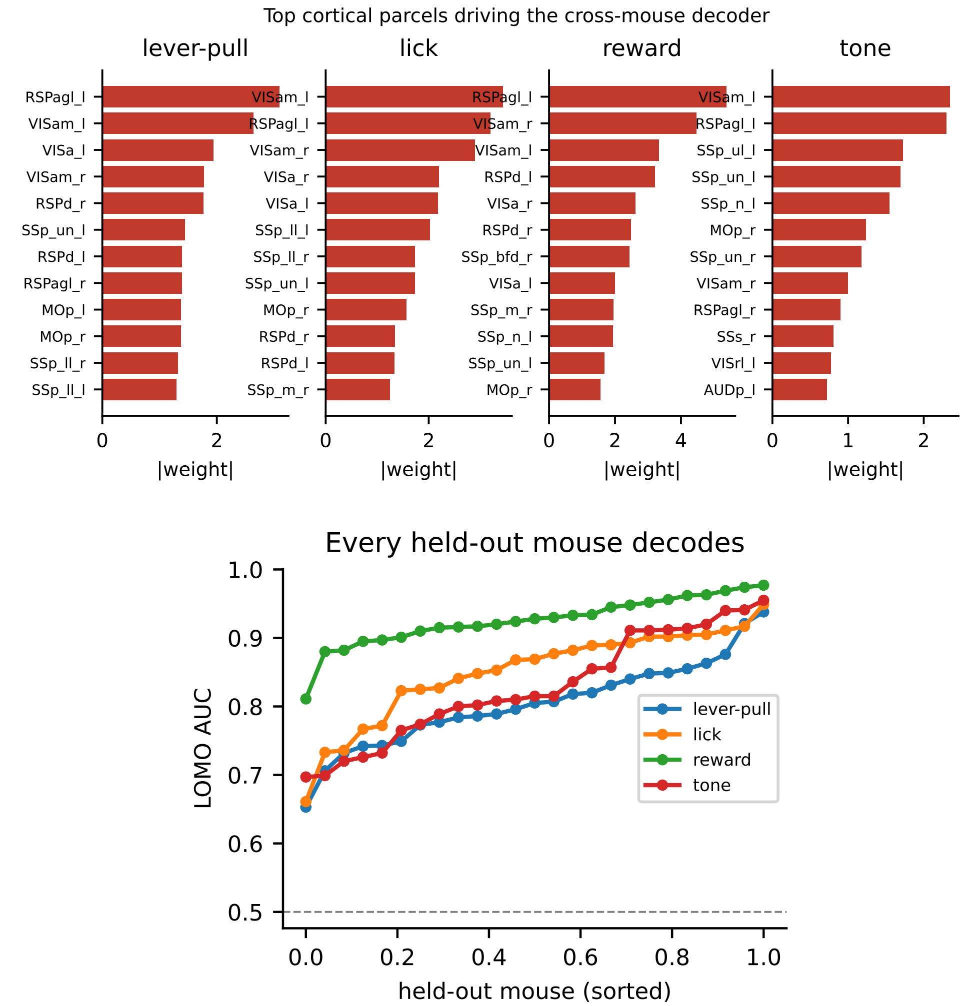

Figure 4: Top: top Allen-atlas parcels by decoder weight—a shared midline retrosplenial/visual-association backbone with target-specific somatosensory/motor modulation. Bottom: per-held-out-mouse LOMO AUC (sorted); every held-out mouse decodes above chance.

6 Conclusion

A strict leave-one-mouse-out analysis in a shared 44-parcel atlas space shows that cortex-wide widefield representations of four behavioral events are read out across unseen mice within 0.03–0.04 AUC of the within-animal ceiling, and that for every event this small cross-subject gap is stable across a two-week protocol and does not widen with decoder capacity, with an independent-cohort replication. We release code and a streamed-feature pipeline that reproduces every number without downloading raw imaging.

Reproducibility statement

All data are open (BraiDyn-BC, DANDI:001425, CC-BY; Cardin–Higley, figshare 175317). We release the full analysis code and a streaming feature pipeline: every BraiDyn-BC table and figure is reproduced from the public parcellated traces without downloading raw widefield movies, and the committed result files (braidyn_4targets.json, braidyn_drift.json, braidyn_nonlinear.json, cardin_results.json) carry every reported value together with per-mouse breakdowns and the 44-parcel ROI table. The independent-cohort replication of

Section 4.4 is the one analysis that is not streamed: it requires the Cardin–Higley parcel-trace archive (six animals, 18 MB) to be fetched from figshare first. All splits, permutation seeds, and model hyperparameters are specified in Section 3 and the released code.

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